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Automatic packaging dispensing system

Abstract

An automatic packaging dispensing system comprises a plurality of different packaging boxes disposed in vertical stacks upon a plurality of packaging box platforms, and a plurality of different wrapping paper packaging sheets disposed within trays housed within a housing. A vacuum pickup device is located at vacuum pickup station toward which a predeterminedly selected one of the plurality of different packaging boxes, or a predeterminedly selected one of the plurality of different wrapping paper packaging sheets, may be moved such that the vacuum pickup device may pick up the predeterminedly selected one of the different packaging boxes or wrapping paper packaging sheets and transfer the predeterminedly selected one of the different packaging boxes or wrapping paper packaging sheets to a downstream location.

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Background/Summary

REFERENCE TO CORRESPONDING APPLICATION (1) This United States Non-Provisional patent application is a Non-Provisional Conversion of U.S. Provisional Patent Application No. 62/817,007 which was filed on Mar. 12, 2019, the priority benefits of which are hereby claimed.

FIELD OF THE INVENTION

(1) The present invention relates generally to food service equipment, and more particularly to a new and improved automatic packaging dispensing system which is to be utilized in restaurants, particularly fast-food restaurants, in order to quickly and accurately prepare food products, such as, for example, conventional hamburger sandwiches which require two buns, or Big Mac® sandwiches which require three buns, or other food items which desire or require to have the food items to be placed within specific packaging boxes or wrapping paper packaging in accordance with specific patron orders.

BACKGROUND OF THE INVENTION

(2) Conventionally, when preparing specific food orders, particularly in a fast-food restaurant, and more particularly, in connection with hamburgers, Big Macs®, Whoppers®, nachos, chicken

nuggets, fish filets, or the like, food preparation personnel must, for example, conventionally utilize different types of packaging for the particular food items, whether such packaging may be, for example, a particular box or a particular wrapping paper packaging, or sometimes, both. Obviously, the conventional packaging of food items, within a particular packaging box or wrapping paper packaging, is somewhat tedious, cumbersome and time-consuming in that the food preparation personnel must select a particularly sized packaging box depending upon the food item to be placed in a packaging box, or to similarly select a particularly sized wrapping paper packaging. This process becomes particularly tedious when different orders, ordered in a serial manner by different patrons at a point-of-sale (POS) location, requires different packaging boxes or different wrapping paper packaging which are to be correlated with or matched to the different serially placed orders. (3) In other words, a first order may be for a regular hamburger which requires a first type of wrapping paper packaging, a second order may be for a Big Mac® which requires a particular packaging box, a third order may be for a fish filet which requires a second type of packaging box, a fourth order may be for chicken nuggets which require a third type of packaging box, and a fifth order may be for a salad which requires a fourth type of packaging box. It can therefore be appreciated that different types of wrapping paper packaging or different types of packaging boxes will be required at unpredictable times in accordance with the different orders placed by the different patrons placed at the point-of-sale (POS) location. Such differences in orders, together with the constant requirement for speed in providing the correct food orders to the patrons can also impose a predetermined amount of tension upon food preparation personnel. It can therefore be readily appreciated still further that such food preparation procedures are very inefficient, time consuming, and labor intensive, particularly for fast-food restaurants wherein quick or rapid cooking or toasting food preparation times are required or desirable in order to produce the finalized food items within relatively short service times which effectively define one of the more important and attractive features or characteristics of fast-food establishments.

(4) A need therefore exists in the art for a new and improved piece of food service equipment. An additional need exists in the art for a new and improved piece of food service equipment which is adapted to prepare specific food items. A still additional need exists in the art for a new and improved piece of food service equipment which is adapted to prepare specific food items, such as, for example, hamburgers, Big Macs®, Whoppers®, nachos, fish filets, chicken nuggets, or the like. A yet additional need exists in the art for a new and improved piece of food service equipment which can be utilized in various different food establishments. A still yet additional need exists in the art for a new and improved piece of food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants. A further need exists in the art for a new and improved piece of food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants, wherein quick turnaround time for preparing food item orders is a highly sought-after feature or characteristic. A still further need exists in the art for a new and improved piece of food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants, wherein the equipment is very efficient in connection with the preparation of specific food items ordered by patrons at the point-of-sale (POS) location. A yet further need exists in the art for a new and improved piece of food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants, wherein the equipment is very efficient in connection with the preparation of the food items or products, and can be operated by means of a minimum number of food preparation personnel.

(5) A still yet further need exists in the art for a new and improved piece of food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants, wherein the equipment is very efficient in connection with the preparation of the food items or products, can be easily operated by means of a minimum number of food preparation personnel, and can be utilized to prepare numerous food items in an automatic, serial manner as

opposed to the food items being singly prepared in a one-by-one fashion. A yet still further need exists in the art for a new and improved piece of food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants, wherein numerous types of packaging can effectively be pre-loaded into the equipment, particularly during lull or relatively slow business hours or times, such that the plurality of food items will then be able to be substantially immediately prepared as orders are inputted into the system for patrons at the point-of-sale (POS) location as opposed to being constrained to being processed in a single or one-by-one fashion. An additional need exists in the art for a new and improved piece of food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants, wherein numerous types of packaging can effectively be pre-loaded into the equipment, particularly during lull or relatively slow business hours or times, and then dispensed automatically such that the plurality of food items will then be able to be substantially immediately prepared in accordance with orders that are inputted into the system for patrons at the point-of-sale (POS) location. A still additional need exists in the art for a new and improved piece of food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants, wherein numerous types of packaging can effectively be pre-loaded into the equipment, particularly during lull or relatively slow business hours or times, and then dispensed automatically such that food preparation personnel are not required to select, nor are involved with the selection of, the particular packaging for a particular food item whereby the plurality of food items will then be able to be substantially immediately prepared as orders are inputted into the system for patrons at the point-of-sale (POS) location.

OVERALL OBJECTIVES OF THE PRESENT INVENTION

(6) Therefore, a primary objective of the present invention is to therefore provide a new and improved piece of food service equipment. An additional objective of the present invention is to provide a new and improved piece of food service equipment which is adapted to prepare specific food items. A still additional objective of the present invention is to provide a new and improved piece of food service equipment which is adapted to prepare specific food items, such as, for example, hamburgers, Big Macs®, Whoppers®, nachos, fish filets, chicken nuggets, or the like. A yet additional objective of the present invention is to provide a new and improved piece of food service equipment which can be utilized in various different food establishments. A still yet additional objective of the present invention is to provide a new and improved piece of food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants. A further objective of the present invention is to provide a new and improved piece of food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants, wherein quick turnaround time for preparing food item orders is a highly sought-after feature or characteristic. A still further objective of the present invention is to provide a new and piece of improved food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants, wherein the equipment is very efficient in connection with the preparation of specific food items ordered by patrons at the point-of-sale (POS) location. A yet further objective of the present invention is to provide or a new and improved piece of food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants, wherein the equipment is very efficient in connection with the preparation of the food items or products, and can be operated by means of a minimum number of food preparation personnel.

(7) A still yet further objective of the present invention is to provide a new and improved piece of food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants, wherein the equipment is very efficient in connection with the preparation of the food items or products, can be easily operated by means of a minimum number of food preparation personnel, and can be utilized to prepare numerous food items in an automatic, serial manner as opposed to the food items being singly prepared in a one-by-one fashion. A yet

still further objective of the present invention is to provide a new and improved piece of food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants, wherein numerous types of packaging can effectively be pre-loaded into the equipment, particularly during lull or relatively slow business hours or times, such that the plurality of food items will then be able to be substantially immediately prepared as orders are inputted into the system for patrons at the point-of-sale (POS) location as opposed to being constrained to being processed in a single or one-by-one fashion. An additional objective of the present invention is to provide a new and improved piece of food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants, wherein numerous types of packaging can effectively be pre-loaded into the equipment, particularly during lull or relatively slow business hours or times, and then dispensed automatically such that the plurality of food items will then be able to be substantially immediately prepared as orders are inputted into the system for patrons at the point-of-sale (POS) location. A still additional objective of the present invention is to provide for a new and improved piece of food service equipment which can be utilized in various different food establishments, such as, for example, fast food restaurants, wherein numerous types of packaging can effectively be pre-loaded into the equipment, particularly during lull or relatively slow business hours or times, and then dispensed automatically such that food preparation personnel are not required to select, nor are involved with the selection of, the particular packaging for a particular food item whereby the plurality of food items will then be able to be substantially immediately prepared as orders are inputted into the system for patrons at the point-of-sale (POS) location.

SUMMARY OF THE INVENTION

(8) The foregoing and other objectives of the present invention are achieved in accordance with the principles and teachings of the present invention through the provision of a new and improved automatic food item packaging dispensing system wherein the system is adapted to dispense both packaging boxes and wrapping paper packaging. The packaging boxes and wrapping paper packaging are located at particular locations along a food preparation line where a vacuum suction cup gripping assembly is utilized to pick up an uppermost one of a plurality of different packaging boxes units, or an uppermost one of a plurality of different wrapping paper packaging units, and to deposit the same onto a conveyor belt so that the packaging boxes or wrapping paper packaging may be moved to the next food preparation station of the food preparation line. The packaging box sub-system comprises a plurality of different packaging boxes that are disposed within different packaging box assemblies which are mounted upon a mounting plate which is moved laterally by means of a linear belt drive so as to selectively move a particular one of the packaging box assemblies to a particular location at which the vacuum suction cups can pick up the uppermost one of the packaging boxes disposed within that particular packaging box assembly.

(9) In turn, each packaging box assembly comprises a box support platform, a peripheral framework for effectively enclosing the boxes upon the platform, a pair of vertically oriented guide rods which serve to retain the packaging boxes in both forward/backward and lateral directions within the peripheral framework, a plurality of inwardly extending flexible fingers to effectively maintain the supply of boxes upon the platform so that together there is defined a magazine supply of boxes, and a packaging box sensor which is aligned with an aperture defined within the box support platform so as to detect whether any boxes remain upon the platform or that the supply of boxes has been depleted. Constant force springs are also operatively connected to the box support platform so as to always bias the box support platform in a vertically upward direction such that the uppermost packaging box is always located at the same location relative to the vacuum suction cups so as to enable the vacuum suction cups to pick up the uppermost packaging box and transfer it to one of the food preparation line conveyors. As the vacuum suction cups remove the uppermost box, the flexible fingers will effectively be moved out of the way so as to permit the uppermost box to be extracted from the magazine supply of boxes but will immediately return to their original

unflexed position so as to prevent a second box from being removed from the magazine along with the uppermost box.

(10) The wrapping paper packaging sub-system comprises a plurality of trays, each of which holds a predetermined number of wrapping paper packaging sheets, wherein the plurality of trays are disposed within a housing which is vertically movable along vertically oriented ball screw actuator driven by means of a suitable motor, wherein the vertically oriented ball screw actuator is also fixedly mounted upon a vertically oriented mounting platform or wall. Each tray is also slidable upon horizontally oriented ball-bearing slides or tracks, which are similar to the mounting of drawers within desks or other furniture, so that the trays can be easily and readily moved from their normal positions internally within the tray housing to an external position relative to the tray housing when packaging sheets within a particular tray needs to be accessed by means of the vacuum suction cups. Each forward end of each wrapping paper tray is provided with a horizontally oriented finger, and a horizontally oriented tray extension cylinder is mounted upon the vertically oriented mounting platform or wall so as to encounter the finger of a particular wrapping paper tray and thereby cause the tray to be moved out from the interior of the paper tray housing to an external position at which the vacuum suction cups can engage the uppermost wrapping paper sheet within that wrapping paper tray and then transfer that wrapping paper sheet to the food preparation line conveyor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various other features and attendant advantages of the present invention will be more fully appreciated from the following detailed description when considered in connection with the accompanying drawings in which like reference characters designate like or corresponding parts throughout the several views, and wherein:

(2) FIG. 1 is a front, top perspective view showing the overall automatic packaging dispensing system constructed in accordance with the principles and teachings of the present invention;

(3) FIG. 2 is a front perspective view showing the packaging box sub-system comprising the plurality of packaging box assemblies which are fixedly secured upon a mounting plate which, in turn, is fixedly secured to a linear belt drive such that the plurality of packaging box assemblies can be moved transversely upon guide rails in order to properly position one of the packaging box assemblies beneath the vacuum suction cup gripping system;

(4) FIG. 3 is a rear perspective view of the mounting plate of FIG. 2 illustrating how the mounting plate is fixedly connected to the motor-driven linear belt drive;

(5) FIG. 4 is a perspective view illustrating the various components comprising one of the packaging box assemblies, including the vertically oriented guide rods which will engage side portions of the packaging boxes so as to maintain the packaging boxes stationary within the packaging box assemblies both in longitudinal or forward/backward as well as lateral or side-to-side modes;

(6) FIG. 4a is an enlarged perspective view of the upper ends of the guide rods as mounted within the peripheral framework of the packaging box assemblies so as to permit the guide rods to be adjusted laterally with respect to each other so as to be capable of accommodating different packaging boxes having different width dimensions;

(7) FIG. 5 is a plan view of a typical packaging box showing the corner interconnections between the upper and lower half sections of the packaging box which is where the vertically oriented guide rods of FIG. 4 will be disposed so as to properly engage the packaging boxes;

(8) FIG. 6 is a side perspective view of the vacuum suction cup gripping assembly illustrating the vertically oriented servo drive motor for moving the plurality of suction cups vertically upwardly

and downwardly;

(9) FIG. 7 is an enlarged perspective view of the vacuum suction cup gripping assembly as illustrated within FIG. 6 illustrating the mounting of the vertically oriented servo drive motor upon a mounting plate which is fixedly connected to a horizontally oriented servo drive motor which is utilized for moving the vacuum suction cup gripping assembly in forward and backward modes so as to properly position the vacuum suction cup gripping assembly over a packaging box or a wrapping paper packaging sheet disposed within one of the trays of the wrapping paper packaging sub-system;

(10) FIG. 8 is a perspective view of the wrapping paper packaging sub-system illustrating the plurality of horizontally oriented, vertically spaced paper trays mounted within a wrapping paper packaging housing and respectively containing a plurality of different wrapping paper packaging sheets;

(11) FIG. 9 is an enlarged front, right side, perspective view illustrating how the wrapping paper packaging housing is fixedly connected to a vertically oriented actuator whereby actuation of the actuator, the wrapping paper packaging housing is moved upwardly and downwardly;

(12) FIG. 10 is rear view of the wrapping paper packaging sub-system illustrating the mounting of the wrapping paper packaging housing upon vertically oriented guide rods fixedly secured to a rear wall of the sub-system, as well as the tray extension cylinder/actuator for moving a selected one of the plurality of wrapping paper trays outwardly from the wrapping paper packaging housing or for returning the selected tray back into the wrapping paper packaging housing once a particular wrapping paper packaging sheet has been removed from the selected tray;

(13) FIG. 11 is a top plan view of one of the wrapping paper packaging trays illustrating additional details as to how the wrapping paper packaging trays are mounted upon the vertically oriented guide rails for vertical movement; and

(14) FIG. 12 is a perspective view of one of the wrapping paper packaging trays illustrating adjustable positioning guides disposed within the wrapping paper packaging tray such that any one of the plurality of wrapping paper packaging trays can accommodate any particularly sized wrapping paper packaging sheets, as well as an aperture which is formed within a relatively central portion of the wrapping paper packaging tray for cooperating with a sensor to indicate that the supply of wrapping paper packaging sheets has been depleted.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

(15) With reference now being made to FIGS. 1-5, the new and improved automatic packaging dispensing system of the present invention is disclosed and is generally indicated by the reference character **100**. More particularly, it is seen that the system **100** is adapted to dispense both packaging boxes and wrapping paper packaging and therefore comprises a packaging box sub-system **102** and a wrapping paper packaging sub-system **104**. The packaging box sub-system **102** comprises a plurality of different types or sizes of packaging boxes that are disposed within different packaging box assemblies **106** which are fixedly mounted at their upper end portions upon a mounting plate **108** which is operatively connected to a linear belt drive **110** by means of a mounting bracket **112** disposed atop a linear slide block **114** which is connected to the linear belt drive **110**. A reversible motor **116** is operatively connected to the linear belt drive **110** so that when the reversible motor **116** is actuated in a first mode for a predetermined amount of time, the linear belt drive **110** will move the mounting plate **108** and the plurality of packaging box assemblies **106** mounted thereon a predetermined distance in a first direction, whereas, when the reversible motor **116** is actuated in a second mode for a predetermined amount of time, the linear belt drive **110** will move the mounting plate **108** and the plurality of packaging box assemblies **106** mounted thereon a predetermined distance in a second opposite direction.

(16) The mounting plate **108** undergoes its lateral, reciprocating movements as a result of being movably attached to, and supported upon, a horizontally oriented guide rail **118** as can be seen in FIGS. 1 and 2. In this manner, any one of the plurality of packaging box assemblies **106** can be

moved so as to be properly positioned relative to, and beneath, a vacuum suction cup pickup and transfer assembly **120**, which can be seen in FIG. **1** and which will be more fully discussed hereinafter, such that the vacuum suction cup pickup and transfer assembly **120** can remove the uppermost packaging box, one exemplary box **122** being disclosed within FIG. **5**, disposed within the particular packaging box assembly **106** and thereby transfer the packaging box **122** to a food preparation line conveyor **124**, as seen in FIG. **1**, such that the particular packaging box may be moved to the next food preparation station of a food preparation line **126**.

(17) With reference now being made to FIG. **4**, one of the packaging box assemblies **106** will be described. More particularly, it is seen that each packaging box assembly **106** comprises a box support platform **128** upon which a plurality of a particular kind or type of packaging boxes **122** are to be stacked so as to effectively form a packaging box magazine from which the uppermost packaging box **122** will be removed by means of the vacuum suction cup assembly **120**. The box support platform **128** actually comprises a horizontally oriented component **130**, upon which the stack of packaging boxes **122** are actually disposed and supported, and an upstanding bracket **132** to which a pair of laterally spaced constant force springs **134** are adapted to be attached although the springs **134** are not shown attached in FIG. **4**. The springs **134** are mounted upon a horizontal spindle **136**, the ends of which are mounted within laterally spaced end brackets **138**. The end brackets **138,138**, in turn, are connected to a peripheral framework **140** which is mounted upon the mounting plate **108** and to which are attached a plurality, for example, four, leaf-spring type fingers **142**, as well as a pair of vertically oriented packaging box guide rods **144** which depend vertically downwardly from opposite sides of the peripheral framework **140**.

(18) The peripheral framework **140** and the vertical guide rods **144** are adapted to maintain the stack of packaging boxes **122** upon the stacking platform component **130**, with the vertical guide rods **144** engaging corner regions **146** of the packaging boxes where are effectively defined between the upper or cover portion of the packaging box **122** and the bottom portion of the packaging box **122**. The disposition of the vertical guide rods **144** within the corner regions **146** of the packaging boxes **122** effectively prevents the plurality of packaging boxes **122** from moving forwardly and backwardly as well as laterally from side to side. Furthermore, with reference being additionally made to FIG. **4a**, it is seen that the upper end portions **141** of the vertical guide rods **144** are bent at an angle of 90° and are seated within slots **143** defined within the peripheral framework **140**. Accordingly, the upper end portions **141** of the pair of vertical guide rods **144** may be moved within the slots **143** and relative to each other such that the distance between the pair of vertical guide rods **144** is rendered adjustable. In this manner, different packaging boxes **122**, having different width dimensions, can be housed or accommodated within the packaging box assembly **106**. It is additionally noted that mounting plates **145** are fixedly secured atop the peripheral framework **140** so as to securely fix the bent end portions **141** of the vertical rods **144** within the slots **143** of the peripheral framework so that, in turn, the pair of vertical rods **144** will remain at their relative desired spacing so as to house or accommodate differently sized packaging boxes **122**.

(19) Accordingly, it can be appreciated that as a stack of packaging boxes **122** are mounted upon the horizontal component **130** of the box support platform **128**, the box support platform **128** will be moved downwardly and the constant force springs **134**, when attached to the rear upstanding wall component **132** of the box support platform **128**, will be extended such that the constant force springs **134** will continuously tend to move the box support platform **128** back toward its uppermost position so as to continuously move the box support platform **128** incrementally upwardly as the uppermost packaging box **122** is removed from the box support platform **128** by means of the vacuum suction cup assembly **120**. The box support platform **128** moves vertically downwardly and upwardly along a vertically oriented guide rail **146** which is fixedly secured at its lower end to a mounting bracket **148**. The plurality of fingers **142** extending into the central part of the peripheral framework **140** supply a sufficient retaining force to the stack of packaging boxes

122 disposed upon the horizontal component **130** of the packaging box platform **128**, however, when the uppermost packaging box **122** is removed by means of the vacuum suction cup assembly **120**, the plurality of fingers will flex upwardly, permitting the uppermost packaging box **122** to be withdrawn from the stack of packaging boxes **122** disposed upon the horizontal component **130** of the packaging box platform **128**, and will effectively slide along the external surface portions of the uppermost packaging box **122** being withdrawn by means of the vacuum suction cup assembly **130** until the fingers **142** snap back to their original positions, thereby preventing any further packaging boxes **122** from being removed from the packaging box assembly **106**. It is lastly noted that the packaging box assembly **106** also includes a packaging box sensor **152** which is adapted to be coaxially aligned with an aperture **154** defined within the horizontal component **130** of the packaging box platform **128**. In this manner, in a manner similar to a photocopying machine warning the user that the supply of paper in the paper tray has been depleted, the sensor **152** will sense when the last packaging box **122** has been removed from the packaging box assembly **106** whereby the supply of packaging boxes **122** needs to be replenished.

(20) Continuing further, and with particular reference being made to FIGS. **6** and **7**, the vacuum suction cup pickup and transfer system **120** will now be described. A pair of suction cups **154** are fixedly attached to the bottom surfaces of a pair of vacuum manifolds **156** so that the system comprises four suction cups **154**. Each one of the vacuum manifolds **156** is fluidically connected to a vacuum generator **158**, and the vacuum generators **158,158** are fixedly mounted upon a horizontally oriented mounting bracket **160**. A plurality of vertically oriented tubes or guide rods **162** have their lower ends fixedly connected to the mounting bracket **160** and are operatively associated with a servo unit **164** which is fixedly mounted upon a vertically extending housing **166**. Accordingly, as the plurality of tubes or guide rods **162** move relative to the servo unit **164**, the plurality of suction cups **154** will accordingly be moved upwardly and downwardly. It is to be noted at this juncture that the control of all movable components, the operation of all valving, not necessarily shown, and the like, comprising the automatic packaging dispensing system **100** are under the control of a suitable control unit which may be, for example, a programmable logic controller **168** schematically illustrated within FIG. **1**, which causes the various operative components to act as described in accordance with patron order data transmitted thereto from the point-of-sale (POS) location.

(21) Continuing further, it is also seen that the vertically oriented housing **166** is fixedly mounted upon a mounting plate **170** and that the mounting plate **170** is fixedly connected to a servo unit **172** which, in turn, is slidably movable upon a linear slide **174**. Accordingly, when the servo unit **172** moves along the linear slide **174**, it will cause the suction cup assembly, comprising the mounting plate **170**, the housing **166**, the tubes **162** and the servo unit **164**, and the mounting bracket **160**, the vacuum generators **158**, the vacuum manifolds **156**, and the vacuum suction cups **154** to move laterally toward the left or right as viewed within FIGS. **6** and **7**, or to move into or out of the page as viewed within FIG. **1**. In a similar manner, the servo unit **164** can effectively cause the tubes or guide rods **162** to move upwardly and downwardly so as to, in turn, cause the mounting bracket **160**, the vacuum generators **158**, the vacuum manifolds **156**, and the vacuum suction cups **154** to move upwardly and downwardly. Accordingly, when these compound movements are properly controlled and timed by means of, for example, the programmable logic controller (PLC) **168**, the vacuum pickup and transfer assembly **120** can be moved to a position whereby the plurality of vacuum suction cups **154** will be disposed above a selected one of the plurality of packaging box assemblies **106** which has been moved into position beneath the vacuum suction cups **154** of the vacuum pickup and transfer assembly **120** by means of the linear belt drive **110**, the vacuum control generators **158** can then be actuated, and then the servo unit **164** can be actuated so as to move the vacuum suction cups **154** downwardly such that the vacuum suction cups **154** can remove or extract the uppermost one of the packaging boxes **122** from the selected one of the plurality of packaging box assemblies **106**. The servo unit **164** can then be actuated so as to elevate the

packaging box **122** that has been removed or extracted from the selected packaging box assembly **106**, the servo unit **172** can then be actuated so as to effectively retract the vacuum suction cup assembly from its forwardmost position above the selected one of the plurality of packaging box assemblies **106** to a predetermined rearwardmost position at which the vacuum suction cups **154** will now be disposed above the food preparation line conveyor **124**. The servo unit **164** can then be actuated again so as to lower the vacuum suction cups **154** toward the food preparation line conveyor **124**, and when the vacuum suction cups **154** are disposed immediately above the food preparation line conveyor **124**, the vacuum generators **158** can be deactivated whereby the packaging box **122** will be released from the vacuum suction cups **154** and deposited onto the food preparation line conveyor **124**.

(22) With reference now being made to FIGS. **8-12**, the wrapping paper packaging sub-system **104** will now be described. More particularly, it is seen that the wrapping paper packaging sub-system **104** comprises a wrapping paper packaging housing **176** within which there is disposed a plurality of horizontally oriented, vertically spaced trays **178**, each one of the trays **178** holding a predetermined number of wrapping paper packaging sheets, and wherein each tray **178** holds a different type or size of wrapping paper packaging sheet. An L-shaped angle iron or angle bracket **182** has its long leg fixedly secured to the rear wall **183** of the housing **176** while the short leg of the L-shaped angle iron or angle bracket **182** is fixedly secured to a slide block **180** which is part of a vertically oriented linear ballscrew actuator **184** which is driven by means of a suitable motor **186**, all of which can best be seen in FIG. **9**. In addition, it is seen that the vertically oriented ball screw actuator **184** is also fixedly mounted upon a vertically oriented mounting platform or wall **188** which is also shown transparently within FIG. **10** for clarity purposes.

(23) As can best be seen in FIG. **11**, the rear wall **183** of the housing **176** is integrally connected to a left wall **190** of the housing **176** from which project a pair of vertically oriented, laterally spaced female track members **192** which are adapted to engage and slide along a pair of vertically oriented, laterally spaced male guide rails **194**. In this manner, as the wrapping paper packaging housing **176** is moved vertically upwardly and downwardly by means of the linear ballscrew actuator **184**, the movement and support of the wrapping paper packaging housing **176** is provided by means of the operatively cooperative track members **192** and guide rails **194**. It is lastly noted, as can also best be seen in FIGS. **10** and **11**, that a vertically oriented rail **193** is mounted upon the left wall **190** of the housing **176**, while a pair of pneumatic brake components **195,195** are mounted upon the mounting platform or wall **188** so as to be disposed upon opposite sides of the vertically oriented rail **193**. These pneumatic brake components **195,195** are provided, and are adapted to be moved toward each other so as to fixedly grasp opposite sides of the vertically oriented rail **193**, so as to prevent the entire wrapping paper packaging housing **176** from falling, for example, power to the motor **186** and the linear ballscrew actuator **184** should be terminated for some reason, such as, for example, a power switch or circuit breaker has been tripped, or the like.

(24) With reference still being made to FIG. **11**, it is further seen that each tray **178** is also slidable upon horizontally oriented ball-bearing slides or tracks **196** which extend from the rear wall **183** of the housing **176** toward the front of the housing **176**, which is open so as to permit the front edge portions of the plurality of trays **178** to project outwardly therefrom, so that the trays **178** can be easily and readily moved from their normal positions disposed internally within the tray housing **176** to external positions relative to the tray housing **176** when packaging sheets within a particular one of the trays **178** need to be accessed by means of the vacuum suction cups **154** of the vacuum pickup and transfer assembly **120**. Each forward end of each wrapping paper tray **178** is provided with a horizontally oriented finger **198**, and a horizontally oriented tray extension piston-cylinder assembly **200** is mounted upon the external surface of the vertically oriented mounting platform or wall **188** such that when the piston of the piston-cylinder assembly **200** is extended, the piston will encounter the finger **198** of a particular wrapping paper tray **178** and thereby cause that particular tray **178** to be moved out from the interior of the paper tray housing **176** to an external position at

which the vacuum suction cups **154** of the vacuum pickup and transfer assembly **120** can engage the uppermost wrapping paper sheet within that particular wrapping paper tray **178** and then transfer that wrapping paper sheet to the food preparation line conveyor **124**.

(25) More particularly, it is to be noted that the distal end of the piston of the piston-cylinder assembly **200** is provided with a head portion **202** which has a slotted section **204**, which can be seen in FIG. **11**, through which the various fingers **198** of the various paper trays **178** can pass as the paper tray housing **176** is moved upwardly or downwardly, however, when the paper tray housing **176** has been moved upwardly or downwardly such that a particular one of the paper trays **178** is located at the correct position for that particular tray **178** to be moved out from the housing **176** such that the wrapping paper packaging sheets disposed within that tray **176** can be accessed, the finger **198** of that particular paper tray **178** will be disposed within the slotted section **204** defined within the head portion **202** of the piston of the piston-cylinder assembly **200** such that when the piston-cylinder assembly **200** is actuated, the head portion **202** of the piston of the piston-cylinder assembly **200** can act upon the finger **198** of that particular paper tray **178** so as to move that particular paper tray **178** outwardly from the paper tray housing **176** whereby the uppermost wrapping paper packaging sheet disposed within that extended tray **178** can now be grasped by means of the vacuum suction cups **154** of the vacuum pickup and transfer assembly **120**.

(26) With reference again being made to FIG. **1**, a wrapping paper packaging sheet sensor **206** is mounted at a location upon the food preparation line **126** that is adjacent to and disposed above the wrapping paper packaging housing **176** such that when a selected paper tray **178** of the wrapping paper packaging housing **176** is extended outwardly from the wrapping paper packaging housing **176** such that the uppermost wrapping paper packaging sheet disposed within the selected paper tray **178** can in fact be grasped by means of the vacuum suction cups **154** of the vacuum pickup and transfer assembly **120** and transferred to the food preparation line conveyor **124**, the wrapping paper packaging sheet sensor **206** can determine the vertical distance to the uppermost wrapping paper packaging sheet disposed within the selected paper tray **178**. It is to be appreciated that this distance will vary as the supply of wrapping paper disposed within a particular paper tray **178** is steadily depleted. Accordingly, the sensor **206** will transmit a signal to the vacuum pickup and transfer assembly **120** such that, for example, the tubes and servo system **162,164** will know precisely how far to move the plurality of suction cups **154** downwardly in order to properly engage and extract the uppermost wrapping paper sheet disposed within the extended paper tray **178**.

(27) Lastly, as can best be appreciated from FIGS. **11** and **12**, each paper tray **178** is seen to have a substantially rectangular configuration so as to house or accommodate the various different wrapping paper packaging sheets. Accordingly, in order to permit any one paper tray **178** to house or accommodate any particularly sized wrapping paper packaging sheet, a pair of adjustable guides **208,210** are respectively slidably mounted within a pair of guide slots **212,214**, all located upon two sides of each paper tray **178** which form a corner section **216** of the paper tray **178**. Set screws **218,220**, or the like, are mounted within the adjustable guides **208,210** so as to fix the adjustable guides **208,210** at desired positions according to the particularly sized wrapping paper packaging sheets to be accommodated or housed the particular paper tray **178**. Such guides **208,210**, guide slots **212,214**, and set screws **218,220** are similar to those components conventionally utilized within photocopying machines for accommodating or housing differently sized photocopy paper sheets such as, for example, letter sized paper or A4 international sized paper. Still yet further, each paper tray **178** is provided with a through-hole **222** to permit, for example, a light beam or other sensor signal to pass therethrough so as to indicate that the supply of wrapper paper packaging sheets, disposed within the tray **178**, has been depleted, such a sensing system again being similar in operation to a paper supply sensor system incorporated within a photocopying machine.

(28) Obviously, many variations and modifications of the present invention are possible in light of the above teachings. For example, in lieu of, or in conjunction with, the Point-of-Sale (POS)

location, the overall system **100** can also include a user interface where patrons can enter their orders whereupon the rest of the system **100**, as controlled by means of the programmable logic controller (PLC) **168**, will automatically dispense the properly required food item packaging. Still further, a brake can be operatively associated with the box support platform **130** so as to ensure that the support platform **130** is fixedly held against the vertically oriented guide rail **146** at a predeterminedly elevated position, the brake being released when the constant force springs **134** are to move the platform **130** upwardly toward the suction cups **154**. The brake effectively works in combination with vacuum sensors which will effectively confirm that sufficient vacuum is being provided to the suction cups **154** and that the platform **130** is to be maintained at its predetermined elevation so as to enable the suction cups **154** to remove the uppermost packaging box **122**. It is therefore to be understood that within the scope of the appended claims, the present invention may be practiced otherwise than as specifically described herein.

REFERENCE NUMBER KEY

(29) **100**—Automatic packaging dispensing system **102**—Packaging box sub-system **104**—Wrapping paper packaging sub-system **106**—Packaging box assemblies **108**—Mounting plate **110**—Linear belt drive **112**—Mounting bracket **114**—Linear slide block **116**—Reversible motor **118**—Horizontally oriented guide rail **120**—Vacuum suction cup pickup and transfer assembly **122**—Exemplary packaging box **124**—Food preparation line conveyor **126**—Food preparation line **128**—Box support platform **130**—Horizontally oriented component of **128** **132**—Upstanding bracket **132** of **128** **134**—Constant force springs **136**—Horizontal spindle **138**—End brackets **140**—Peripheral framework **141**—Upper end portions of guide rods **144** **142**—Leaf-spring fingers **143**—Slots defined within peripheral framework **140** **144**—Vertically oriented packaging box guide rods **145**—Mounting plates securing ends **141** of guide rods **144** within slots **143** **146**—Corner regions of packaging box **122** **148**—Mounting bracket **152**—Packaging box sensor **154**—Suction cups **156**—Vacuum manifolds **158**—Vacuum generators **160**—Mounting bracket **162**—Vertically oriented guide rods **164**—Servo unit **166**—Housing **168**—Programmable logic controller (PLC) **170**—Mounting plate **172**—Servo unit **174**—Linear slide **176**—Wrapping paper packaging housing **178**—Trays of housing **176** **180**—Slide block on linear ball screw actuator **184** **182**—Angle bracket **183**—Rear wall of housing **176** **184**—Linear ballscrew actuator **186**—Motor drive for actuator **184** **188**—Mounting platform **188** **190**—Left wall of housing **176** **192**—Track members **193**—Vertically oriented rail **194**—Guide rails for track members **192** **195**—Pneumatic brake components **196**—Ball bearing slide tracks for trays **178** **198**—Finger on forward end of each tray **178** **200**—Piston-cylinder assembly **202**—Head portion of piston-cylinder assembly **200** **204**—Slotted section of head portion **202** **206**—Wrapping paper packaging sheet sensor **208,210**—Adjustable guides in each tray **178** **212,214**—Slots for guides **208,210** **216**—Corner section of tray **178** **218,220**—Set screws for guides **208,210** **222**—Through hole within tray **178** for detecting depletion of wrapping sheets

Claims

1. An automatic packaging dispensing system for automatically dispensing packaging, disposed at a pickup location, to a downstream location, the system comprising: a plurality of packaging materials, within which items can be packaged, disposed within at least one vertically stacked array at said pickup location; a pickup device, disposed at said pickup location, for picking up an uppermost one of said plurality of packaging materials, wherein said pickup device is operatively mounted upon at least one vertically oriented guide rod which is operatively associated with a first motor drive for moving said pickup device along with said at least one vertically oriented guide rod such that said pickup device can move vertically upwardly and downwardly, and wherein said at least one vertically oriented guide rod and said first motor drive are operatively mounted upon a horizontally oriented slide mechanism which is operatively associated with a second motor drive

for moving said at least one vertically oriented guide rod, said first motor drive, and said pickup device along said horizontally oriented slide mechanism such that said pickup device can move horizontally forwardly and backwardly; and a control system for: moving said plurality of packaging materials toward said pickup location such that said uppermost one of said plurality of packaging materials, disposed within said at least one vertically stacked array, is disposed at said pickup location; moving said pickup device downwardly along with said at least one vertically oriented guide rod and into a position to pick up said uppermost one of said plurality of packaging materials disposed within said at least one vertically stacked array; moving said pickup device upwardly along with said at least one vertically oriented guide rod so as to remove said uppermost one of said plurality of packaging materials from remaining ones of said plurality of packaging materials disposed within said at least one vertically stacked array; and moving said pickup device, said at least one vertically oriented guide rod, and said first motor drive horizontally away from said pickup location, and along said horizontally oriented slide mechanism, so as to transfer said uppermost one of said plurality of packaging materials from said pickup location toward the downstream location.

2. The system as set forth in claim 1, wherein: said plurality of packaging materials comprises a plurality of packaging boxes.

3. The system as set forth in claim 2, wherein: said plurality of packaging boxes are different and are disposed within different vertically stacked arrays of said at least one vertically stacked array.

4. The system as set forth in claim 3, wherein: said vertically stacked arrays of packaging boxes are disposed upon different packaging box support platforms.

5. The system as set forth in claim 4, wherein: each one of said different packaging box support platforms is provided with a through-hole for permitting a sensor signal to pass therethrough and thereby detect when a supply of packaging boxes disposed upon a particular one of said different packaging box support platforms has been depleted.

6. The system as set forth in claim 5, wherein: each one of said packaging box support platforms has a pair of laterally spaced, vertically oriented guide rods operatively associated with said packaging box support platform for maintaining the respective vertically stacked array of packaging boxes upon said packaging box support platform.

7. The system as set forth in claim 6, wherein: said pair of laterally spaced, vertically oriented guide rods are laterally adjustable with respect to said packaging box support platform so as to accommodate differently sized packaging boxes.

8. The system as set forth in claim 4, wherein: at least one constant force spring is operatively connected to each one of said packaging box support platforms so as to constantly bias each one of said packaging box support platforms in an upward direction such that an uppermost one of said plurality of packaging boxes is always disposed at a position which will enable said uppermost one of said plurality of packaging boxes to be picked up by said pickup device.

9. The system as set forth in claim 2, wherein: said pickup device comprises at least one vacuum suction cup for picking up one of said packaging boxes.

10. The system as set forth in claim 1, wherein: said plurality of packaging materials comprises a plurality of wrapping paper packaging sheets.

11. The system as set forth in claim 10, wherein: said plurality of wrapping paper packaging sheets are disposed within a plurality of wrapping paper packaging sheet trays.

12. The system as set forth in claim 11, wherein: said plurality of wrapping paper packaging sheet trays are disposed within a housing; a first actuator mounted at a predetermined location upon said housing for operatively engaging a selected one of said plurality of wrapping paper packaging sheet trays so as to move said selected one of said plurality of wrapping paper packaging sheet trays out from said housing such that said wrapping paper packaging sheets disposed within said selected one of said plurality of wrapping paper packaging sheet trays is available to be picked up by said pickup device; and a second actuator mounted upon said housing for moving said plurality of

wrapping paper packaging sheet trays toward said predetermined location at which said first actuator is located such that said selected one of said plurality of wrapping paper packaging sheet trays will be disposed at a position at which said first actuator can move said selected one of said plurality of wrapping paper packaging sheet trays out from said housing.

13. The system as set forth in claim 12, wherein: said first actuator comprises a horizontally oriented piston-cylinder assembly; and said second actuator comprises a vertically oriented linear ball screw actuator.

14. The system as set forth in claim 11, wherein: each one of said plurality of trays is provided with adjustable guides such that each one of said plurality of trays can accommodate differently sized wrapping paper packaging sheets.

15. The system as set forth in claim 11, wherein: each one of said plurality of trays is provided with a through-hole for permitting a sensor signal to pass therethrough and thereby detect when a supply of wrapping paper packaging sheets disposed within said tray has been depleted.

16. The system as set forth in claim 10, wherein: said pickup device comprises at least one vacuum suction cup for picking up at least one of said wrapping paper packaging sheets.

17. An automatic packaging dispensing system for automatically dispensing packaging, disposed at a pickup location, to be moved to a downstream location, the system comprising: a first set of packaging materials within which a first type of item can be packaged, said first set of packaging materials being disposed within a first vertically stacked array; a second set of packaging materials, different from said first set of packaging materials, within which a second type of item can be packaged, said second set of packaging materials being disposed within a second vertically stacked array; a pickup device, disposed at said pickup location, for picking up an uppermost one of said first set of packaging materials disposed within said first vertically stacked array or an uppermost one of said second set of packaging materials disposed within said second vertically stacked array, wherein said pickup device is operatively mounted upon at least one vertically oriented guide rod which is operatively associated with a first motor drive for moving said pickup device along with said at least one vertically oriented guide rod such that said pickup device can move vertically upwardly and downwardly, and wherein said at least one vertically oriented guide rod and said first motor drive are operatively mounted upon a horizontally oriented slide mechanism which is operatively associated with a second motor drive for moving said at least one vertically oriented guide rod, said first motor drive, and said pickup device along said horizontally oriented slide mechanism such that said pickup device can move horizontally forwardly and backwardly; and a control system for: moving said first set of packaging materials or said second set of packaging materials toward said pickup location such that said uppermost one of said first set of packaging materials or said uppermost one of said second set of packaging materials, disposed within said first or second vertically stacked array, is disposed at said pickup location; moving said pickup device downwardly along with said at least one vertically oriented guide rod and into a position to pick up said uppermost one of said first or second set of packaging materials disposed within said first or second vertically stacked array; moving said pickup device upwardly along with said at least one vertically oriented guide rod so as to remove said uppermost one of said first or second set of packaging materials, disposed within said first or second vertically stacked array, from remaining ones of said first or second set of packaging materials disposed within said first or second vertically stacked array; and moving said pickup device, said at least one vertically oriented guide rod, and said first motor drive horizontally away from said pickup location, and along said horizontally oriented slide mechanism, so as to transfer said uppermost one of said first or second set of packaging materials from said pickup location toward the downstream location.

18. The system as set forth in claim 17, wherein: said first set of packaging materials comprises a set of packaging boxes; and said second set of packaging materials comprises a set of wrapping paper packaging sheets.

19. The system as set forth in claim 17, wherein: said pickup device comprises at least one suction

cup which can selectively pick up a selected one of said first set of packaging materials or can selectively pick up a selected one of said second set of packaging materials.
